Chemical Thermodynamics: Gibbs Free Energy & Reaction Kinetics

1. Spontaneity and Chemical Equilibrium

The thermodynamic criterion for reaction spontaneity at constant temperature and pressure is dictated by the Gibbs Free Energy change:

$$\Delta G = \Delta H - T\Delta S$$

When Delta G < 0, the reaction proceeds exergonically (spontaneously). At dynamic equilibrium, Delta G = 0 and connects to equilibrium constant K:

$$\Delta G^\circ = -RT \ln(K_{eq})$$